

Lecture 6

Kalman filter, observability analysis, KF-based VINS

Kalman filter is a linear estimator based on Gaussian assumption.

Before that, let's first start with an MAP problem:

Given:

$$\begin{cases} x_0 \sim N(\hat{x}_0, P_0) & \rightarrow \text{Prior} \\ x_1 = f(x_0) + w_0, \quad w_0 \sim N(0, Q) & \rightarrow \text{motion model: (IMU)} \\ z_1 = h(x_1) + n_1, \quad n_1 \sim N(0, R) & \rightarrow \text{measurement model (vision)} \end{cases}$$

Target: $x_1 \sim N(\hat{x}_1, P_1)$

How to compute it?

$$\max_x p(x|z) = \max_x \frac{p(z|x)p(x)}{p(z)} = \max_x \frac{p(z_1|x_1, x_0) p(x_1, x_0)}{p(z_1)}$$

$$= \max_x \frac{p(z_1|x_1, x_0) p(x_1|x_0) p(x_0)}{p(z_1)}$$

$\downarrow$ z_1 only depends on x_1 , drop x_0

$$= \max_x \frac{p(z_1|x_1) p(x_1|x_0) p(x_0)}{p(z_1)}$$

$$\Leftrightarrow \min_x \|z_1 - h(x_1)\|_R^2 + \|x_1 - f(x_0)\|_Q^2 + \|x_0 - \hat{x}_0\|_{P_0}^2$$

Linearization:

$$x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}, \text{ linearization: } \hat{x} = \begin{bmatrix} \hat{x}_0 \\ \hat{x}_1 \end{bmatrix}$$

$$\text{then: } x_1 = f(x_0) + w_0$$

$$\Rightarrow \hat{x}_1 + \tilde{x}_1 \approx f(\hat{x}_0) + F(x_0 - \hat{x}_0) + w_0$$

$\triangleq$ linear

$$\hat{x}_1 + \tilde{x}_1 = f(\hat{x}_0) + F \tilde{x}_0 + w_0 \rightarrow x_{1|0} = f(x_0)$$

$$\Rightarrow r_x = \hat{x}_1 - f(\hat{x}_0) = F \tilde{x}_0 - \tilde{x}_1 + w_0 = \begin{bmatrix} F & -I_3 \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix} + w_0$$

$$z_1 = h(x_1) + n_1$$

$$\Rightarrow z_1 = h(\hat{x}_1) + H(x_1 - \hat{x}_1) + n_1$$

$$\Rightarrow \tilde{z}_1 = z_1 - h(\hat{x}_1) = H \tilde{x}_1 + n_1 = \begin{bmatrix} 0 & H \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix} + n_1$$

Then the cost function becomes:

$$\text{cost} = \|\tilde{z} - H \tilde{x}_1\|_R^2 + \|r_x - \begin{bmatrix} F & -I_3 \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix}\|_{Q_0}^2 + \|\tilde{x}_0\|_{P_0}^2$$

solution 1: solve it incrementally:

$$\left\{ \begin{array}{l} \text{step 1: } \|\tilde{x}_0\|_{P_0}^2 + \|r_x - \begin{bmatrix} F & -I_3 \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix}\|_{Q_0}^2 \rightarrow x_{1|0} \sim \mathcal{N}(\hat{x}_{1|0}, P_{1|0}) \\ \text{step 2: } \|\tilde{x}_{1|0}\|_{P_{1|0}}^2 + \|\tilde{z} - H \tilde{x}_1\|_R^2 \rightarrow x_{1|1} \sim \mathcal{N}(\hat{x}_{1|1}, P_{1|1}) \end{array} \right.$$

$$\text{step 1: Prior + motion: } \min_x \|\tilde{x}_0\|_{P_0}^2 + \|r_x - \begin{bmatrix} F & -I_3 \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix}\|_{Q_0}^2$$

Hence: The cost:

$$C = \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix}^T \begin{bmatrix} P_0^{-1} + F^T Q_0^{-1} F & -F^T Q_0^{-1} \\ -Q_0^{-1} F & Q_0^{-1} \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix} - 2 \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix}^T \begin{bmatrix} F^T \\ -I_3 \end{bmatrix} Q_0^{-1} r_x + r_x^T Q_0^{-1} r_x$$

$$\frac{\partial C}{\partial \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix}} = 0 \text{ gives the information system:}$$

$$\begin{bmatrix} P_0^{-1} + F^T Q_0^{-1} F & -F^T Q_0^{-1} \\ -Q_0^{-1} F & Q_0^{-1} \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix} = \begin{bmatrix} F^T Q_0^{-1} r_x \\ -Q_0^{-1} r_x \end{bmatrix}$$

Then, marginalizing x_0 (schur complement onto the x_1 block) and applying the

matrix inversion lemma,

$$[Q_0^{-1} - Q_0^{-1} F (P_0^{-1} + F^T Q_0 F)^{-1} F^T Q_0^{-1}] \tilde{x}_1 = -Q_0^{-1} r_x + Q_0^{-1} F (P_0^{-1} + F^T Q_0 F)^{-1} F^T Q_0^{-1} r_z$$

$$\boxed{(Q_0 + F P_0 F^T)^{-1}} \tilde{x}_1 = - (Q_0 + F P_0 F^T)^{-1} r_x$$

P_{110}^{-1}

Hence:
$$\begin{cases} \tilde{x}_1 = -r_x \\ P_{110} = Q_0 + F P_0 F^T \end{cases} \quad \tilde{x}_1 = -r_x \quad \begin{cases} \text{if } r_x = 0, & \hat{x}_{10} = f(\hat{x}_0) \\ \text{if } r_x \neq 0, & \hat{x}_{10} = f(\hat{x}_0) - r_x \end{cases}$$

step 2:
$$\| \tilde{x}_1 + r_x \|^2_{P_{110}} + \| \tilde{z} - H \tilde{x}_1 \|^2_R$$

$$\begin{aligned} C_2 &= (\tilde{x}_1 + r_x)^T P_{110}^{-1} (\tilde{x}_1 + r_x) + (\tilde{z} - H \tilde{x}_1)^T R^{-1} (\tilde{z} - H \tilde{x}_1) \\ &= \tilde{x}_1^T P_{110}^{-1} \tilde{x}_1 + 2 \tilde{x}_1^T P_{110}^{-1} r_x + r_x^T P_{110}^{-1} r_x + \tilde{z}^T R^{-1} \tilde{z} - 2 \tilde{z}^T R^{-1} H \tilde{x}_1 + \tilde{x}_1^T H^T R^{-1} H \tilde{x}_1 \\ &= \tilde{x}_1^T (P_{110}^{-1} + H^T R^{-1} H) \tilde{x}_1 + 2 \tilde{x}_1^T (P_{110}^{-1} r_x - H^T R^{-1} \tilde{z}) + \tilde{z}^T R^{-1} \tilde{z} + r_x^T P_{110}^{-1} r_x \end{aligned}$$

Hence:
$$\frac{\partial C_2}{\partial [\tilde{x}_0, \tilde{x}_1]} = 0 \Rightarrow \underbrace{(P_{110}^{-1} + H^T R^{-1} H)}_{\Sigma = P_{111}^{-1}} \tilde{x}_1 = H^T R^{-1} \tilde{z} - P_{110}^{-1} r_x$$

Hence:
$$\tilde{x}_1 = (P_{110}^{-1} + H^T R^{-1} H)^{-1} (H^T R^{-1} \tilde{z} - P_{110}^{-1} r_x)$$

$$= \underbrace{(P_{110}^{-1} + H^T R^{-1} H)^{-1} H^T R^{-1}}_{\Downarrow} \tilde{z} - (P_{110}^{-1} + H^T R^{-1} H)^{-1} P_{110}^{-1} r_x$$

$$\begin{aligned} & (P_{110} - P_{110} H^T (H P_{110} H^T + R)^{-1} H P_{110}) H^T R^{-1} \tilde{z} \\ &= (P_{110} H^T R^{-1} - P_{110} H^T (H P_{110} H^T + R)^{-1} H P_{110} H^T R^{-1}) \tilde{z} \\ &= P_{110} H^T (R^{-1} - (H P_{110} H^T + R)^{-1} H P_{110} H^T R^{-1}) \tilde{z} \\ &= P_{110} H^T (H P_{110} H^T + R)^{-1} ((H P_{110} H^T + R) R^{-1} - H P_{110} H^T R^{-1}) \tilde{z} \end{aligned}$$

$$= P_{110} H^T (H P_{110} H^T + R)^{-1} \tilde{z}$$

$$\tilde{x}_1 = P_{110} H^T (H P_{110} H^T + R)^{-1} \tilde{z} - \boxed{(P_{110}^{-1} + H^T R^{-1} H)^{-1} P_{110}^{-1} r_x}$$

Hence: we get:

$$\tilde{x}_1 = P_{110} H^T (H P_{110} H^T + R)^{-1} \tilde{z} - \boxed{(P_{110}^{-1} + H^T R^{-1} H)^{-1} P_{110}^{-1} r_x}$$

if $r_x \rightarrow 0$, then: $\tilde{x}_1 = P_{110} H^T (H P_{110} H^T + R)^{-1} \tilde{z}$

$$P_{111} = (P_{110} + H^T R^{-1} H)^{-1}$$

$$= P_{110}^{-1} - P_{110} H^T (H P_{110} H^T + R)^{-1} H P_{110}$$

Therefore, we have the $x_{111} \sim N(\hat{x}_{111}, P_{111})$.

summary of the process:

$$x_0 \sim N(\hat{x}_0, P_0)$$

Propagation: $x_1 = f(x_0) + w_0$

$$\hat{x}_{1|0} = f(\hat{x}_0), \quad P_{1|0} = F P_0 F^T + Q_0$$

update: $z_1 = h(x_1) + r_1$

$$\begin{cases} \tilde{z}_1 = z_1 - h(\hat{x}_{1|0}) & \tilde{x}_1 = H \tilde{x}_1 + r_1 \\ S_1 = H P_{1|0} H^T + R_1 \end{cases}$$

$$K = P_{1|0} H^T \boxed{(H P_{1|0} H^T + R)^{-1}} \quad \text{Kalman gain}$$

$$\tilde{x}_1 = P_{1|0} H^T (H P_{1|0} H^T + R)^{-1} \tilde{z}_1 \quad (r_x \rightarrow 0)$$

$$\hat{x}_{111} = \hat{x}_{1|0} + \tilde{x}_1 = \hat{x}_{1|0} + P_{1|0} H^T (H P_{1|0} H^T + R)^{-1} \tilde{z}_1$$

$$\begin{cases} P_{11} = P_{10} - P_{10} H^T (H P_{10} H^T + R)^{-1} H P_{10} \end{cases}$$

This is the EKF from MLE/MAP formulation.

Solution 2: Solve by batch method:

$$\text{cost} = \|\tilde{z} - H\tilde{x}\|_R^2 + \|\begin{bmatrix} r_x - [F \quad -I] \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix} \end{bmatrix}\|_{Q_0}^2 + \|\tilde{x}_0\|_{P_0}^2$$

$$\text{The: } \begin{bmatrix} P_0^T + F^T Q_0^T F & -F^T Q_0^T \\ -Q_0^T F & Q_0^T + H^T R^T H \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_1 \end{bmatrix} = \begin{bmatrix} F^T Q_0^T r_x \\ -Q_0^T r_x + H^T R^T \tilde{z} \end{bmatrix}$$

Then::

$$\begin{aligned} [H^T R^T H + (Q_0^T - Q_0^T F (P_0^T + F^T Q_0^T F)^{-1} F^T Q_0^T)] \tilde{x}_1 &= H^T R^T \tilde{z} - Q_0^T r_x + Q_0^T F (P_0^T + F^T Q_0^T F)^{-1} F^T Q_0^T r_x \\ &= H^T R^T \tilde{z} - (Q_0^T - Q_0^T F (P_0^T + F^T Q_0^T F)^{-1} F^T Q_0^T) r_x \end{aligned}$$

If we define: $P_{10} = Q_0^T - Q_0^T F (P_0^T + F^T Q_0^T F)^{-1} F^T Q_0^T$, then:

$$(H^T R^T H + P_{10}) \tilde{x}_1 = H^T R^T \tilde{z} - P_{10} r_x$$

$$\begin{cases} P_{11} = (P_{10}^T + H^T R^T H)^{-1} \\ \tilde{x}_1 = (H^T R^T H + P_{10})^{-1} (H^T R^T \tilde{z} - P_{10} r_x) \end{cases}$$

It's the same as EKF as $r_x \rightarrow 0$

Discussion:

① EKF is a special form for MLE/MAP

... sequentially

② EKF is the way to solve the problem incrementally.

$$\textcircled{3} \quad \left\{ \begin{array}{l} x_k \sim N(\hat{x}_k, P_k) \rightarrow \text{prior} \\ x_{k+1} = f(x_k) + w_k \rightarrow \text{motion, IMU} \\ z_{k+1} = h(x_{k+1}) + n_{k+1} \rightarrow \text{measurements, vision} \end{array} \right\} \text{VINS}$$

④ IMU motion model:

$$\dot{x} = f(x, n_I) \Rightarrow x_{k+1} = f(x_k, n_{I,d}) \quad (\text{lecture 5})$$

vision model:

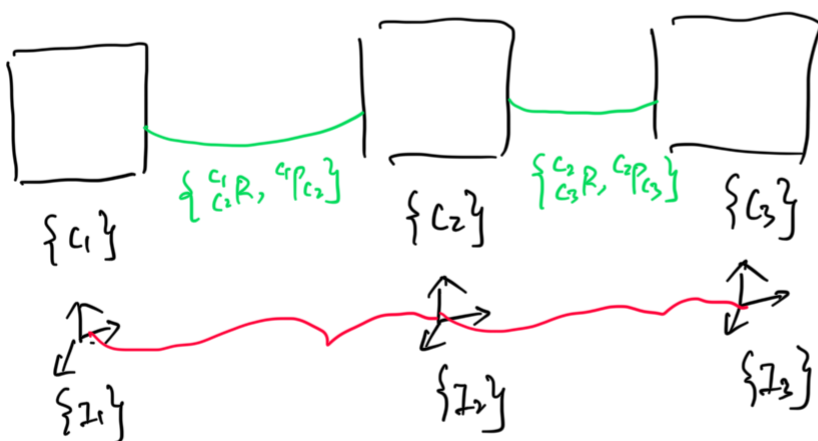
$$z_c = h_c(x) + n_c \quad (\text{lecture 3, 4})$$

⑤ Initialization: in order to run EKF, we need the prior

$$x_k \sim N(x_k, P_k)$$

$$\text{For IMU: } x_I = \left(\begin{matrix} q_R, q_P, q_b, b_g, b_a \end{matrix} \right)$$

we need to estimate q_b, b_g, b_a to start the estimator.



From vision: using the vision SLAM pipeline: $\{C_2R, qP_2\}, \{C_3R, qP_3\}$

From IMU integration:

$$\begin{cases} \mathbf{I}_{k+1}^R = \mathbf{I}_k^R \mathbf{I}_{k+1}^T \mathbf{I}_k^R \\ \mathbf{G}_{\mathbf{P}_{k+1}} = \mathbf{G}_{\mathbf{P}_k} + \mathbf{I}_{\mathbf{V}_k} \delta t + \mathbf{I}_k^R \Delta \mathbf{P}_k + \frac{1}{2} \mathbf{I}_k^R \mathbf{g} \delta t_k^2 \\ \mathbf{I}_{\mathbf{V}_{k+1}} = \mathbf{I}_{\mathbf{V}_k} + \mathbf{I}_k^R \Delta \mathbf{V}_k + \mathbf{I}_k^R \mathbf{g} \delta t_k \end{cases}$$

$$\begin{cases} \mathbf{I}_{\mathbf{P}_{k+1}} = \mathbf{I}_k^R (\mathbf{G}_{\mathbf{P}_{k+1}} - \mathbf{G}_{\mathbf{P}_k}) = \mathbf{I}_k^R \mathbf{V} \delta t_k + \frac{1}{2} \mathbf{I}_k^R \mathbf{g} \delta t_k^2 \\ \mathbf{I}_{\mathbf{P}_{k+2}} = \mathbf{I}_{k+1}^R (\mathbf{G}_{\mathbf{P}_{k+2}} - \mathbf{G}_{\mathbf{P}_{k+1}}) = \mathbf{I}_{k+1}^R \mathbf{V} \delta t_{k+1} + \frac{1}{2} \mathbf{I}_{k+1}^R \mathbf{g} \delta t_{k+1}^2 \\ = \mathbf{I}_{k+1}^R \left(\mathbf{I}_k^R \mathbf{V}_{k+1} \delta t_{k+1} + \frac{1}{2} \mathbf{I}_k^R \mathbf{g} \delta t_{k+1}^2 \right) \end{cases}$$

Hence:

$$\begin{aligned} \begin{bmatrix} \mathbf{I}_k & \mathbf{I}_{\mathbf{P}_{k+1}} \\ \mathbf{I}_{k+1}^R & 1 \end{bmatrix} &= \begin{bmatrix} \mathbf{I}_k^R & \mathbf{I}_k^R \mathbf{P}_k \\ 0 \text{ IFS} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{C}_k^R & \mathbf{C}_k^R \mathbf{P}_{k+1} \\ 0 \text{ IFS} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I}_k^R & \mathbf{P}_2 \\ 0 \text{ IFS} & 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{I}_k^R \mathbf{C}_k^R & \mathbf{I}_k^R \mathbf{C}_k^R \mathbf{P}_{k+1} + \mathbf{I}_k^R \mathbf{P}_k \\ 0 \text{ IFS} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I}_k^R & \mathbf{P}_2 \\ 0 \text{ IFS} & 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{I}_k^R \mathbf{C}_{k+1}^R \mathbf{I}_k^R & \mathbf{I}_k^R \mathbf{C}_{k+1}^R \mathbf{P}_2 + \mathbf{I}_k^R \mathbf{C}_k^R \mathbf{P}_{k+1} + \mathbf{I}_k^R \mathbf{P}_k \\ 0 \text{ IFS} & 1 \end{bmatrix} \end{aligned}$$

Hence:

$$\begin{cases} \mathbf{I}_k^R \mathbf{C}_{k+1}^R \mathbf{P}_2 + \mathbf{I}_k^R \mathbf{C}_k^R \mathbf{P}_{k+1} + \mathbf{I}_k^R \mathbf{P}_k = \mathbf{I}_k^R \mathbf{V} \delta t_k + \frac{1}{2} \mathbf{I}_k^R \mathbf{g} \delta t_k^2 \\ \mathbf{I}_{k+1}^R \mathbf{C}_{k+2}^R \mathbf{P}_2 + \mathbf{I}_{k+1}^R \mathbf{C}_{k+1}^R \mathbf{P}_{k+2} + \mathbf{I}_{k+1}^R \mathbf{P}_k = \mathbf{I}_{k+1}^R \left(\mathbf{I}_k^R \mathbf{V}_{k+1} \delta t_{k+1} + \frac{1}{2} \mathbf{I}_k^R \mathbf{g} \delta t_{k+1}^2 \right) \end{cases}$$

At the same time:

$$\mathbf{I}_k^R \mathbf{V} - \mathbf{I}_k^R \mathbf{V}_{k+1} = \Delta \mathbf{V}_k + \mathbf{I}_k^R \mathbf{g} \delta t_k \Rightarrow$$

$$I_k V - I_k V_{I_{k+1}} - I_k g \delta t_k = \Delta V_k$$

$$\frac{I_{k+1}}{I_k} R (I_k V_{I_{k+1}} - I_k V_{I_{k+2}} - I_k g \delta t_{k+1}) = \Delta V_{k+1}$$

$$\text{Hence: } \begin{bmatrix} -\frac{I_k}{C} R^{C_k} P_{k+1} & \frac{1}{2} \delta t_k^2 I_3 & \delta t_k I_3 & 0_3 \\ -\frac{I_k}{C} R^{C_{k+1}} P_{k+2} & \frac{1}{2} \frac{I_{k+1}}{I_k} R \delta t_{k+1}^2 & 0_3 & \frac{I_{k+1}}{I_k} R \delta t_k \\ 0_{3 \times 1} & -\delta t_k I_3 & I_3 & -I_3 \\ 0_{3 \times 1} & -\frac{I_{k+1}}{I_k} R \delta t_{k+1} I_3 & \frac{I_{k+1}}{I_k} R & -\frac{I_{k+1}}{I_k} R \end{bmatrix} \begin{bmatrix} s \\ I_k g \\ I_k V \\ I_k V_{I_{k+1}} \\ I_k V_{I_{k+2}} \end{bmatrix} = \begin{bmatrix} \frac{I_k}{C} R^{C_k} R^{C_k} P_k + I_k P_k \\ \frac{I_k}{C} R^{C_{k+1}} R^{C_k} P_k + I_k P_k \\ \Delta V_k \\ \Delta V_{k+1} \end{bmatrix}$$

If no scale, s : we need at least 3 frame
 If scale s , we need at least 4 frames